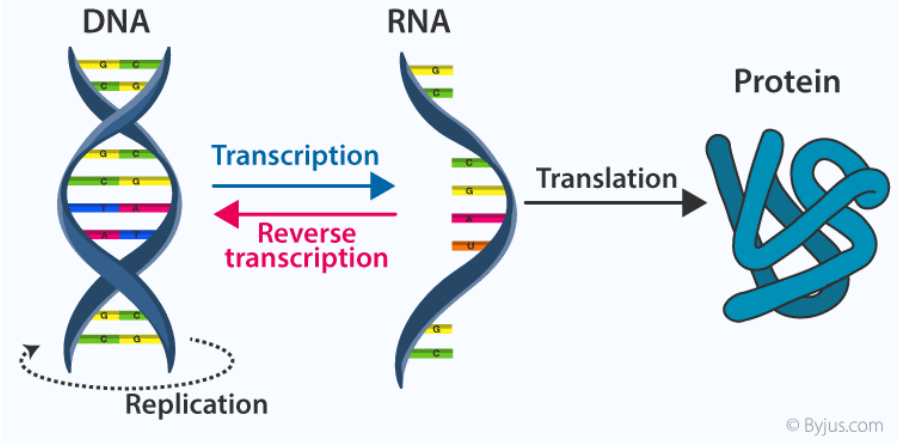
Computational Biology

Session 5

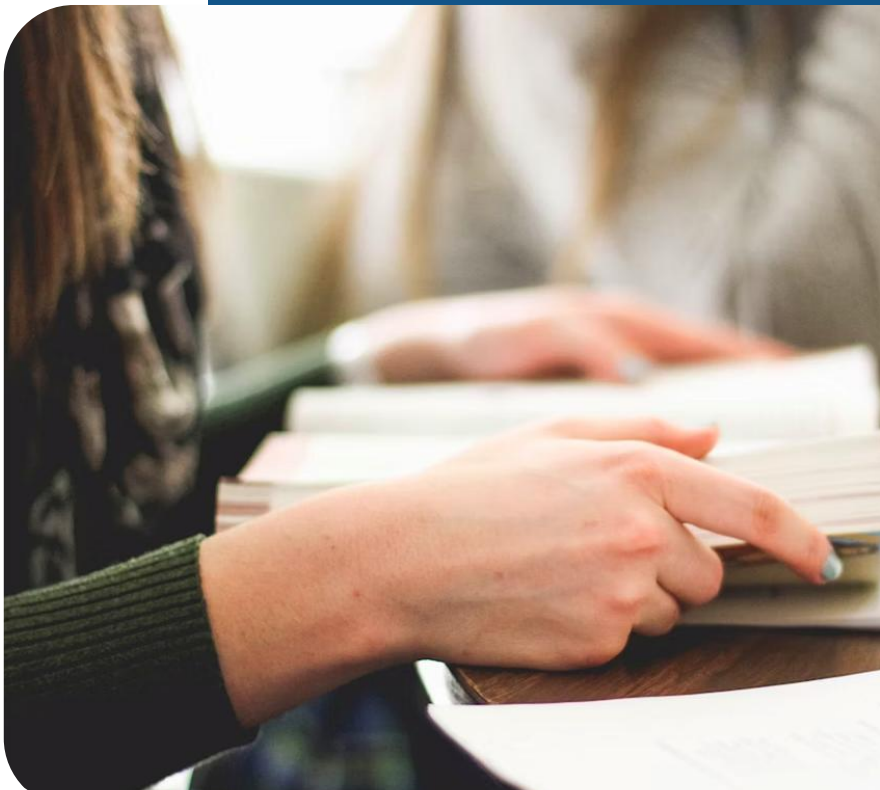


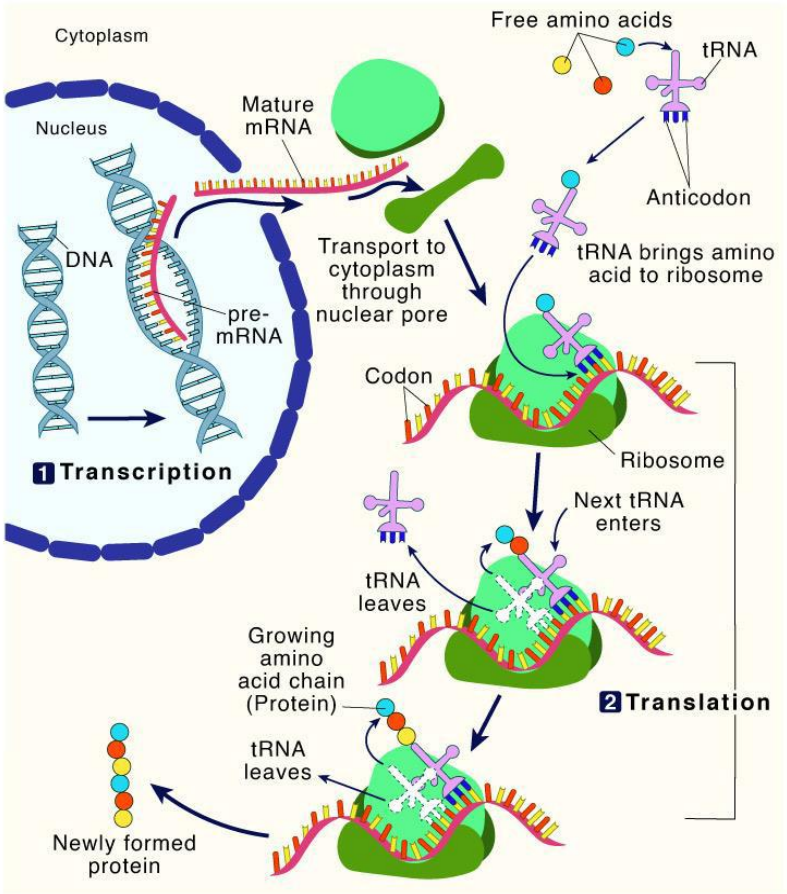
Today's Agenda



- Post **Transcriptional** Modification
- Post **Translational** Modification

Post Transcriptional Modification





What is Post-Transcriptional Modification?

Post-transcriptional modification refers to all molecular processes that occur after transcription (DNA → RNA) but before translation (RNA → protein).

Big Picture Flow

DNA → pre-mRNA → (Processing) → mature mRNA → Protein

Why It Matters

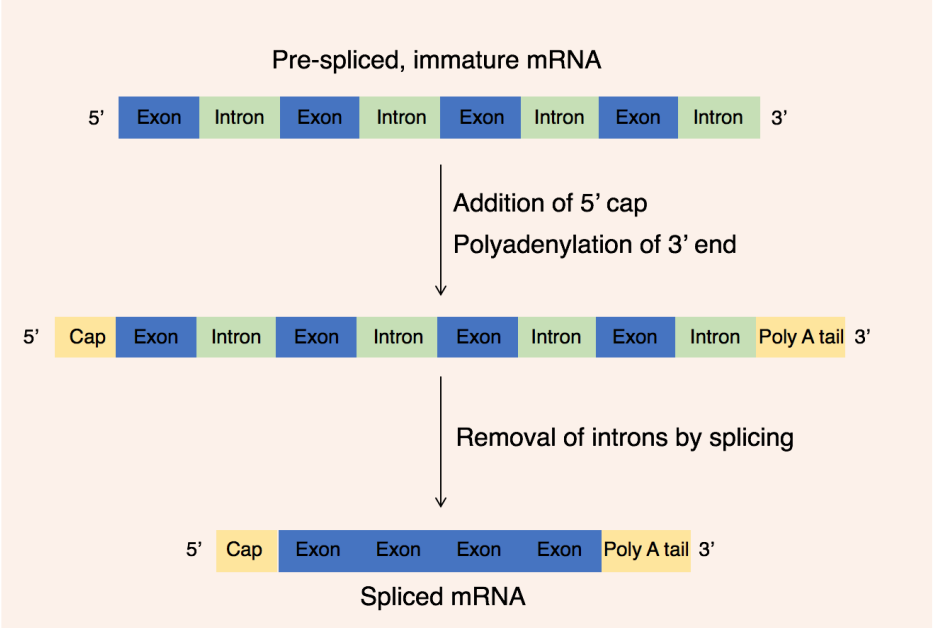
- Ensures RNA stability
- Controls gene expression
- Increases protein diversity
- Enables tissue-specific regulation

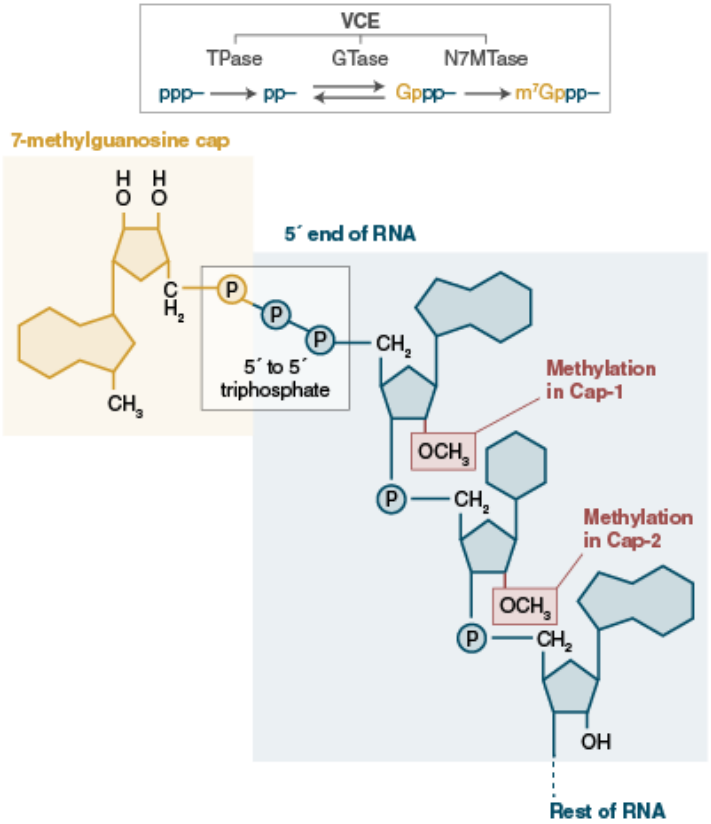
Post-Transcriptional Modification Main Processes

In eukaryotes, three major processes occur:

- 5′ Capping
- RNA Splicing
- 3′ Polyadenylation (Poly-A Tail)

Together, they convert pre-mRNA into mature mRNA.





Post-Transcriptional Modification - 5′ Capping

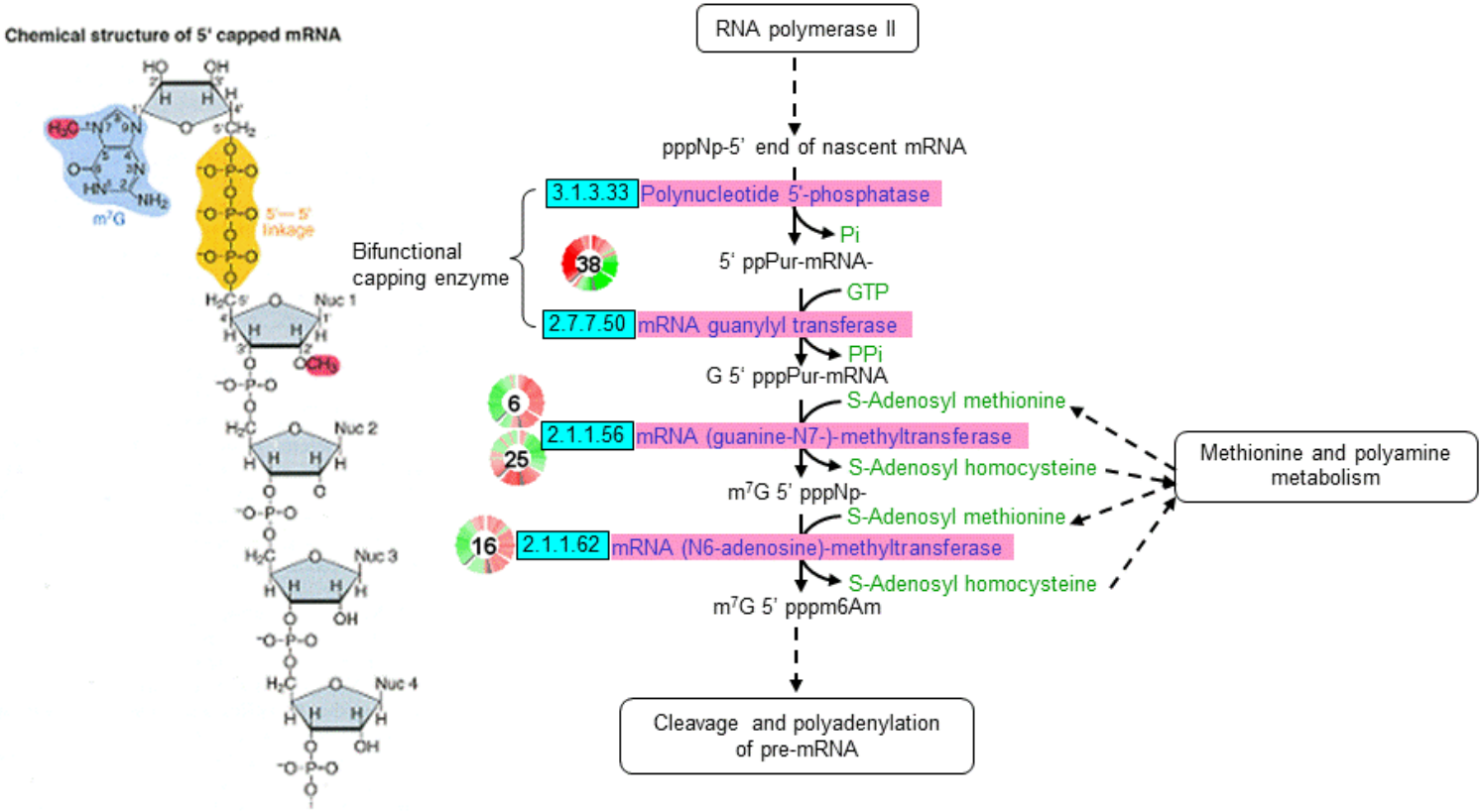
What Happens?
Addition of 7-methylguanosine (m7G) to the 5′ end.

When?
Occurs shortly after transcription begins.

Functions
Protects mRNA from degradation
Required for nuclear export
Required for ribosome recognition
Initiates translation

Without the cap → mRNA is rapidly degraded.

Post-Transcriptional Modification – 5′ Capping



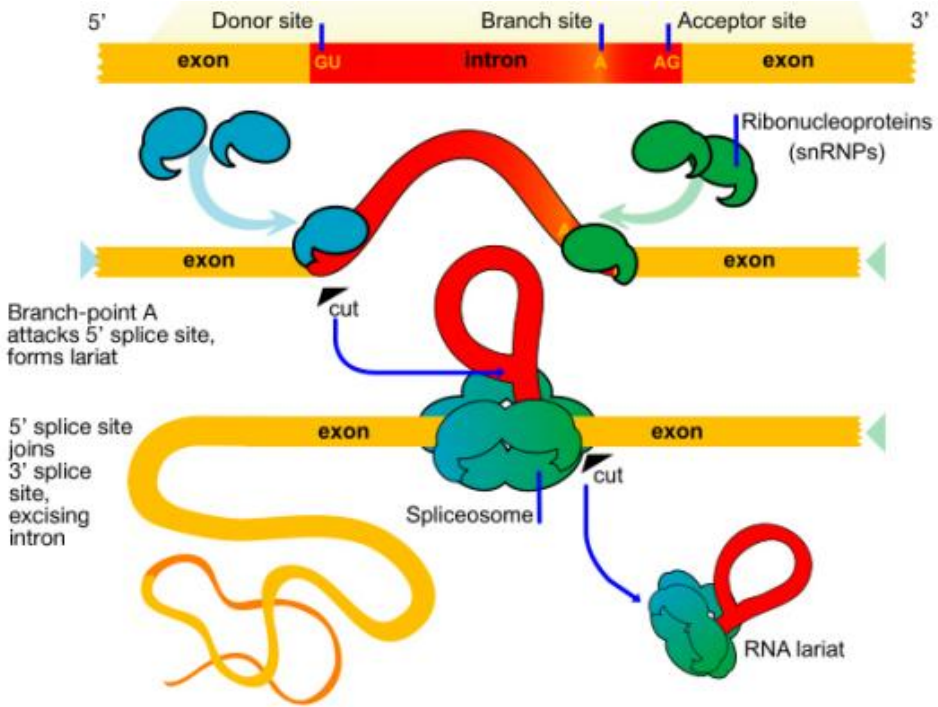
Post-Transcriptional Modification – RNA Splicing

What Happens?
Introns removed
Exons joined together

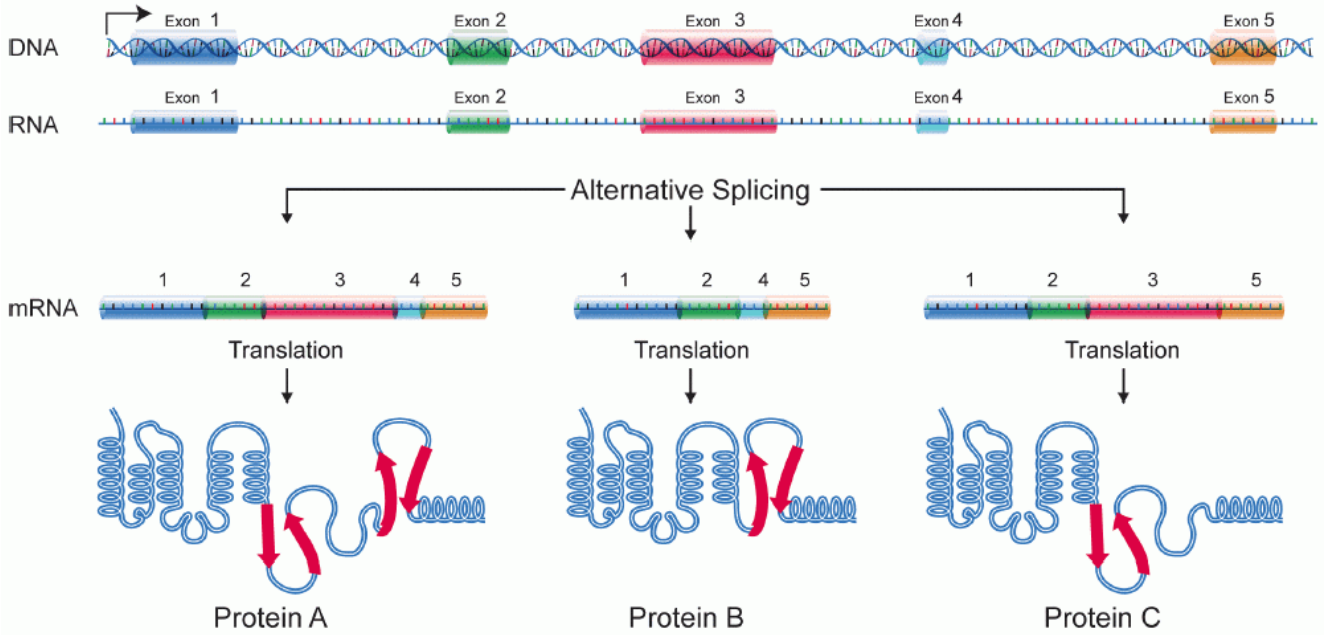
Performed By:
Spliceosome (snRNP complex)

Why Important?
Produces functional coding sequence
Enables alternative splicing
One gene → multiple protein isoforms

Example:
~20,000 human genes → >100,000 proteins



Post-Transcriptional Modification – RNA Splicing



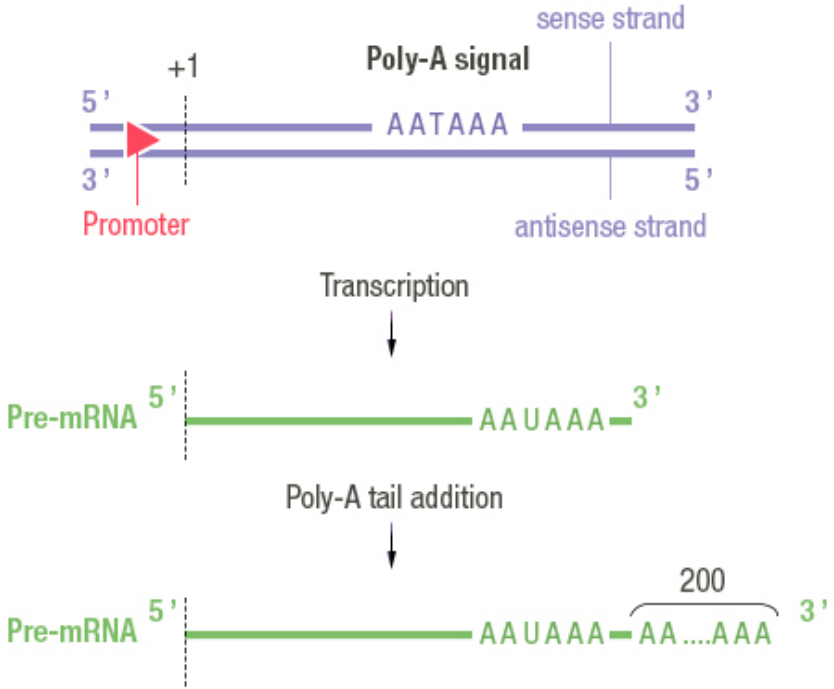
The spliceosome identifies introns by recognizing conserved RNA sequence signals: the **5′ splice site (GU)**, the **branch point adenine**, and the **3′ splice site (AG)**. These sequence motifs recruit small nuclear ribonucleoproteins (snRNPs) that assemble the spliceosome, remove the intron as a lariat, and join the surrounding exons.

Post-Transcriptional Modification – RNA Splicing

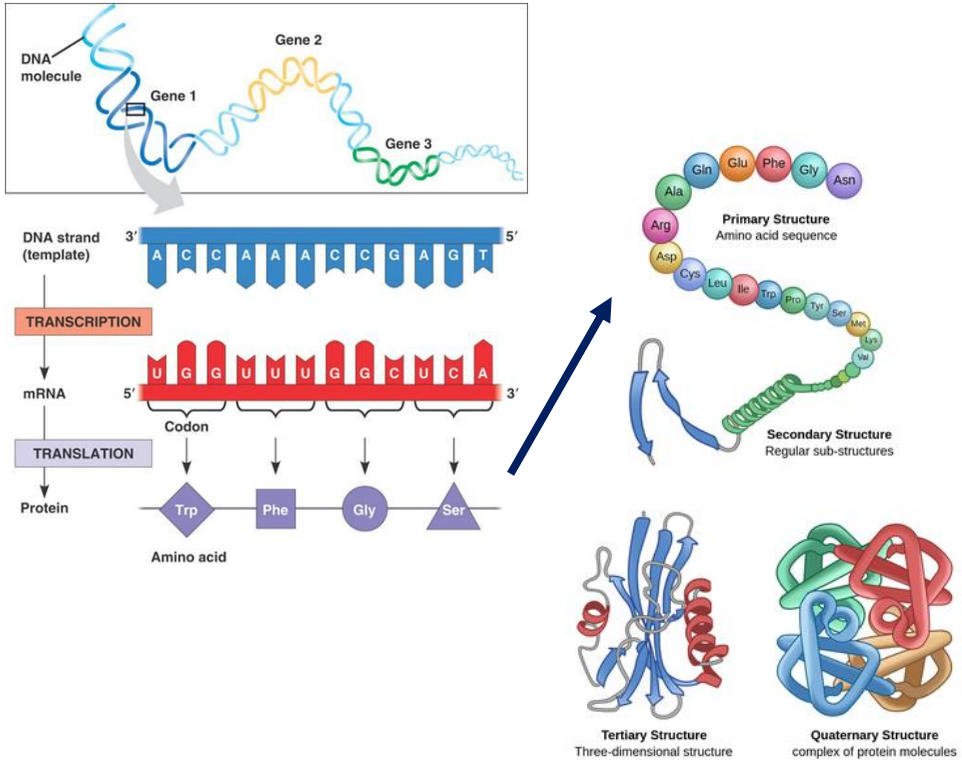
What Happens?
Addition of ~200 Adenine nucleotides at 3′ end.

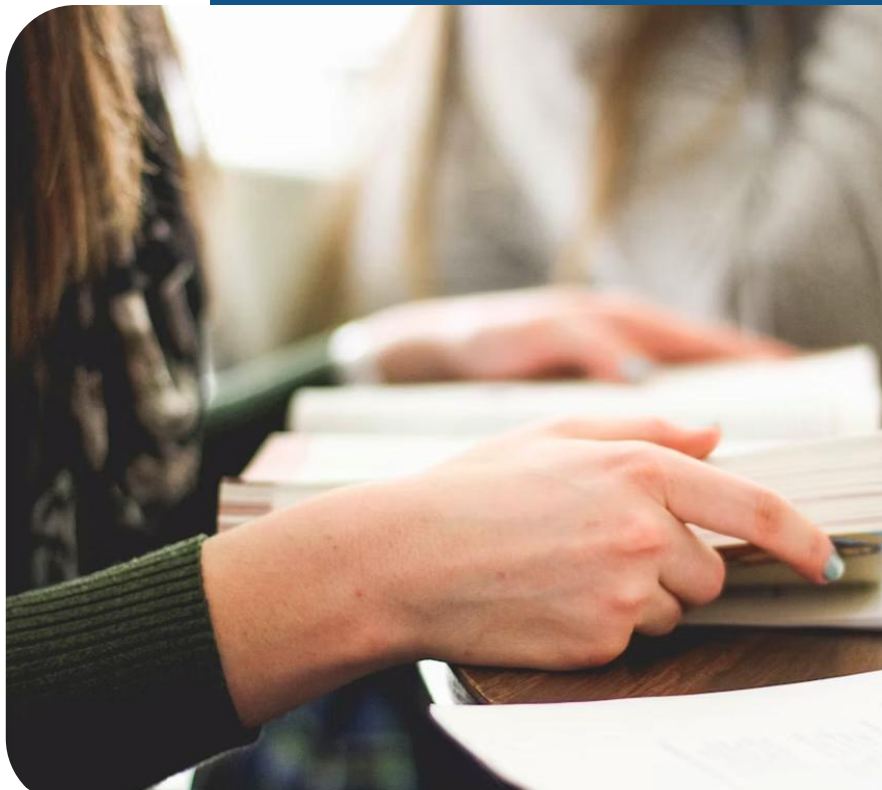
Functions
Increases mRNA stability
Helps nuclear export
Enhances translation efficiency
Regulates mRNA lifespan

Longer poly-A tail → more stable transcript.



Post Translational Modification





Protein Active Site

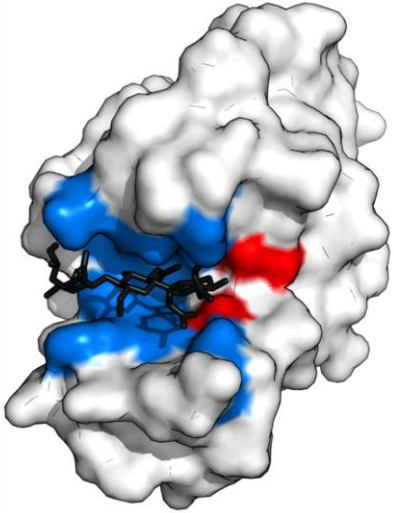
PROTEIN STRUCTURE

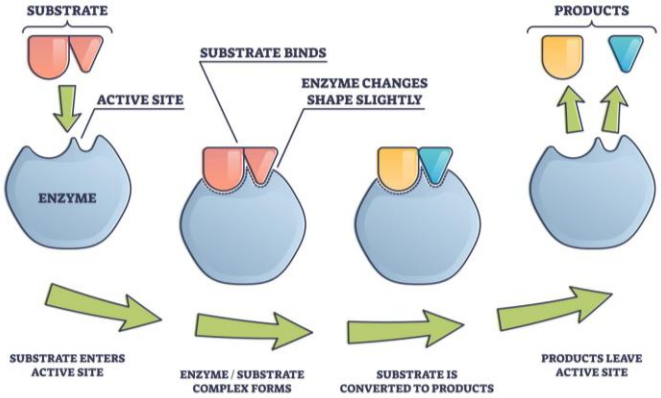
Scaffold to support and position active site

ACTIVE SITE

BINDING SITES
Bind and orient substrate(s)

CATALYTIC SITE
Reduce chemical activation energy





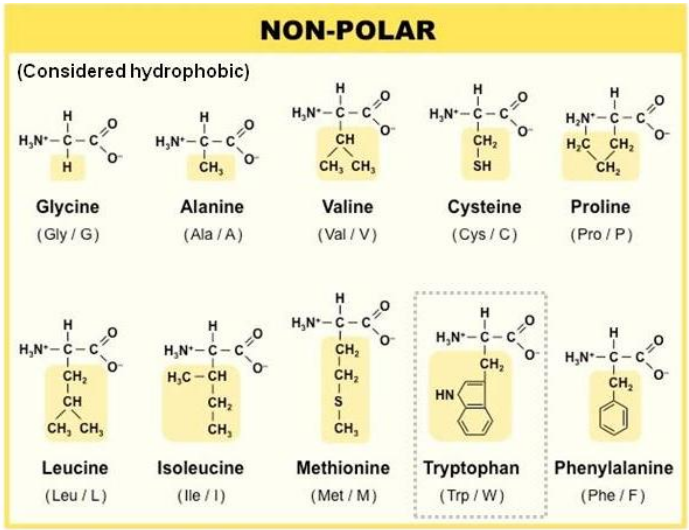
What Is an Active Site?
The active site is a specific 3D region of a protein (usually an enzyme)

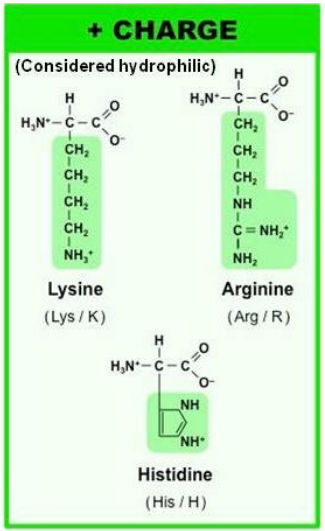
Where?
Substrate binds
Catalysis occurs

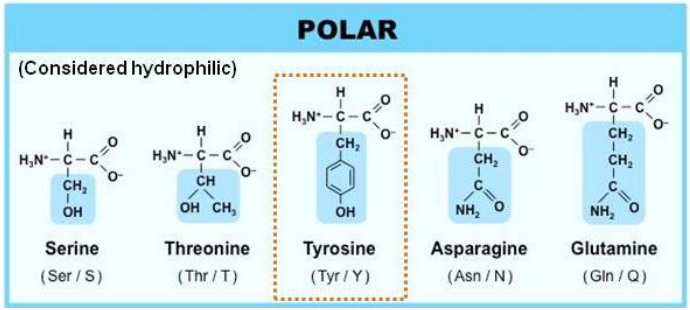
Key Concepts
Formed by precise folding of the protein
Amino acids far apart in sequence come together in 3D
Depends on charge, polarity, and spatial orientation

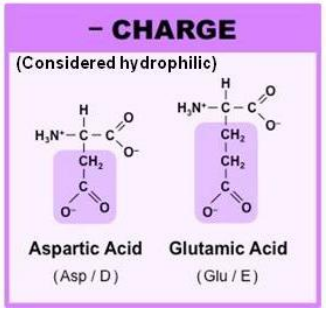
Even a small structural change can drastically alter protein function.

Protein Active Site









What Is an Active Site?
The active site is a specific 3D region of a protein (usually an enzyme)

Where?
Substrate binds
Catalysis occurs

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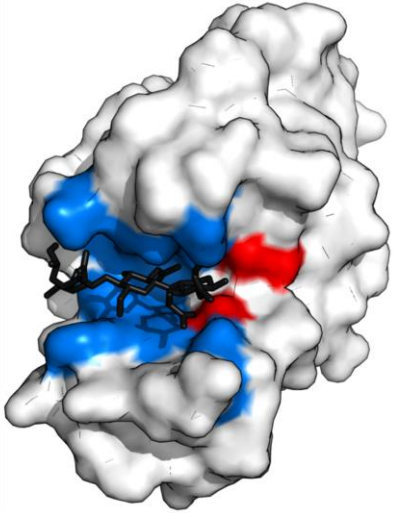
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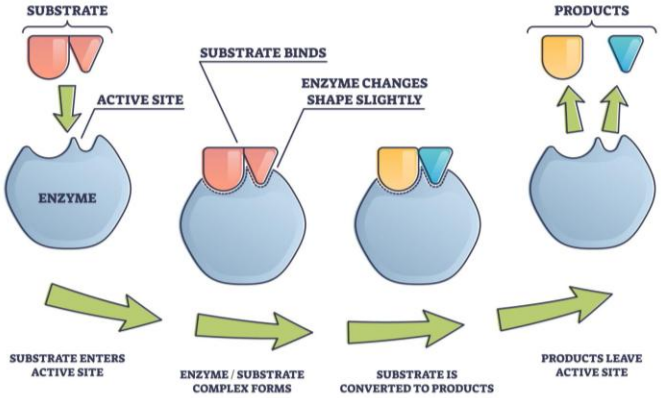
Scaffold to support and position active site

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BINDING SITES
Bind and orient substrate(s)

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Reduce chemical activation energy





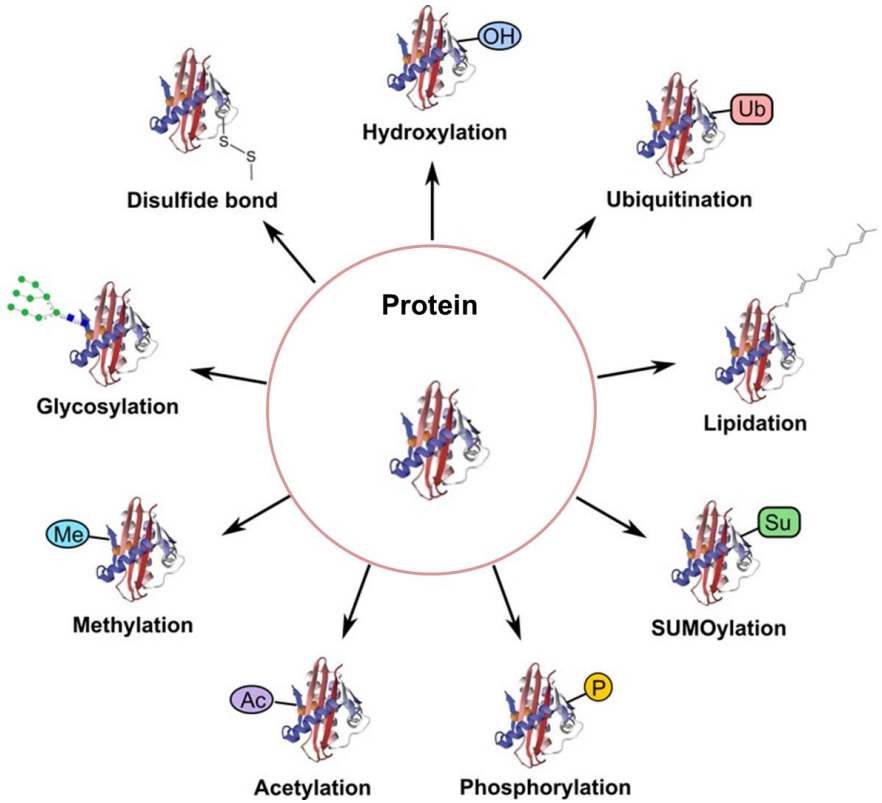
If enzymes are always active:
Metabolism becomes unbalanced
Cell growth becomes uncontrolled
Cellular damage accumulates

Protein function can change through:
Conformational shifts
Protein–protein interaction
Cofactor binding
Chemical modification of amino acids

When specific residues are chemically modified:
Charge changes
Shape shifts
Active site structure alters
Activity turns ON or OFF

This chemical control mechanism is called **Post-Translational Modification (PTM)**

Post-translational modification (PTM)



Definition

Post-translational modification (PTM) refers to chemical modifications made to a protein after translation is complete.

Gene Expression Flow

DNA → RNA → Protein → PTM → Functional Protein

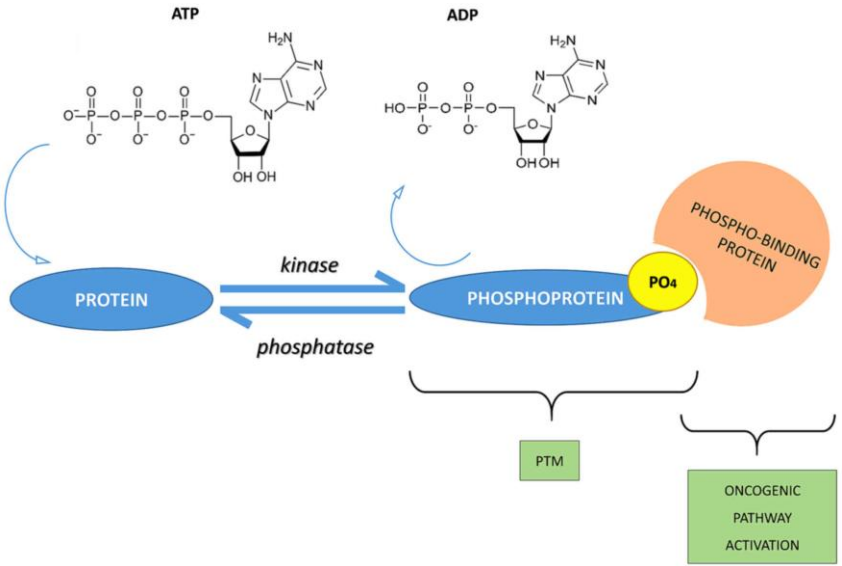
Key Idea

Newly synthesized proteins are often:

1. Inactive
2. Unstable
3. Incorrectly localized

PTMs convert them into their functional form.

Phosphorylation

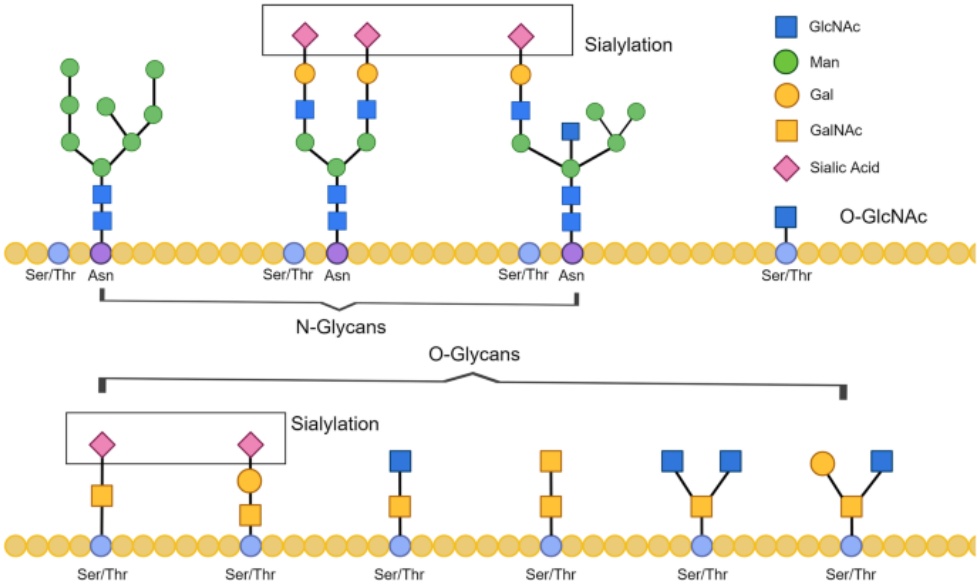


What Happens?
Addition of a phosphate group (PO_4^{3-}) to:
Serine (Ser)
Threonine (Thr)
Tyrosine (Tyr)

Enzymes
Kinase → adds phosphate
Phosphatase → removes phosphate

Function
Turns proteins ON/OFF
Central in cell signalling
Controls growth and metabolism

Glycosylation



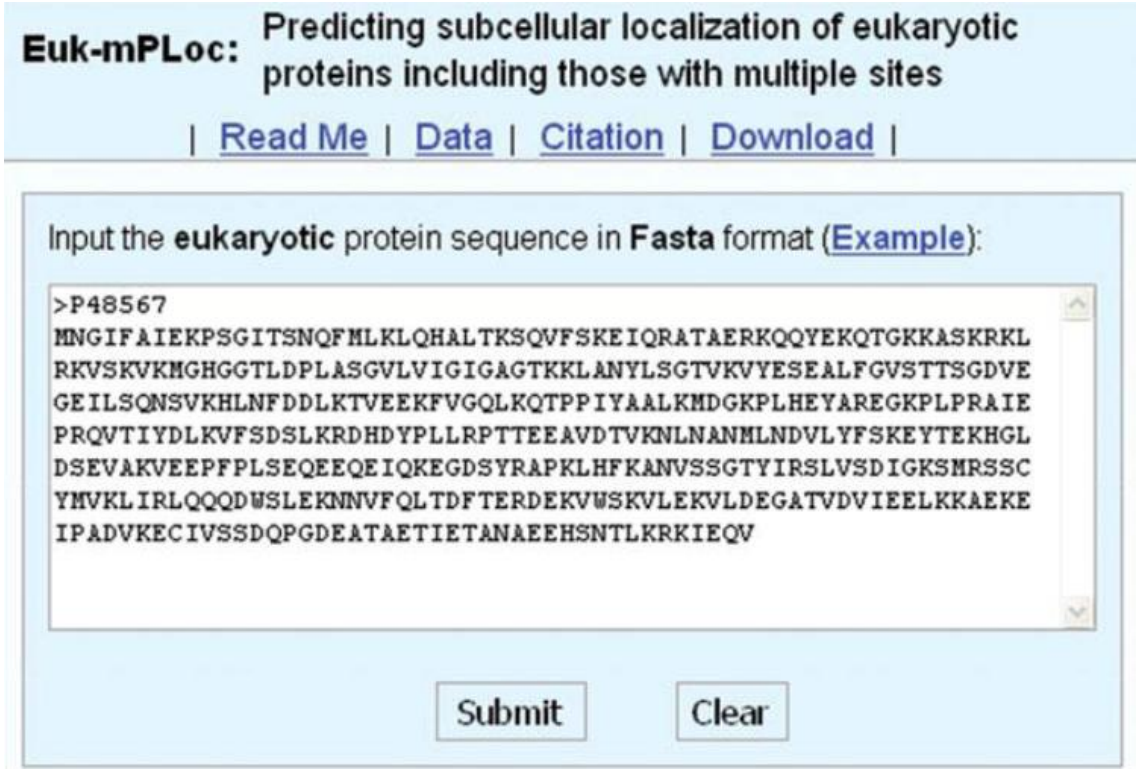
What Happens?
Attachment of sugar molecules to protein.

Types
N-linked (Asparagine)
O-linked (Serine/Threonine)

Function
Protein folding
Stability
Cell recognition
Immune response

Common in membrane and secreted proteins.

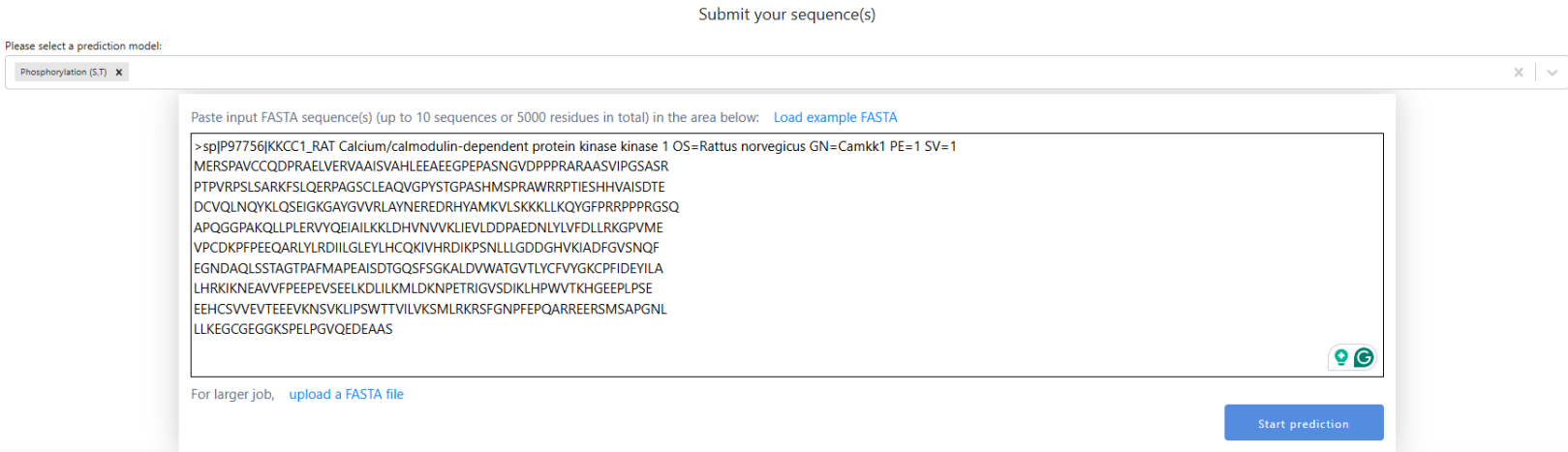
Protein Data Sequence

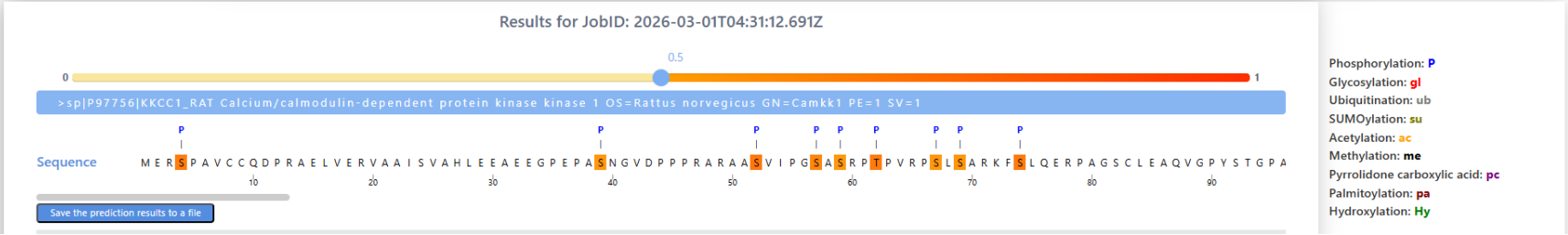


https://www.uniprot.org
https://www.ncbi.nlm.nih.gov/protein/
https://www.rcsb.org
https://www.ensembl.org

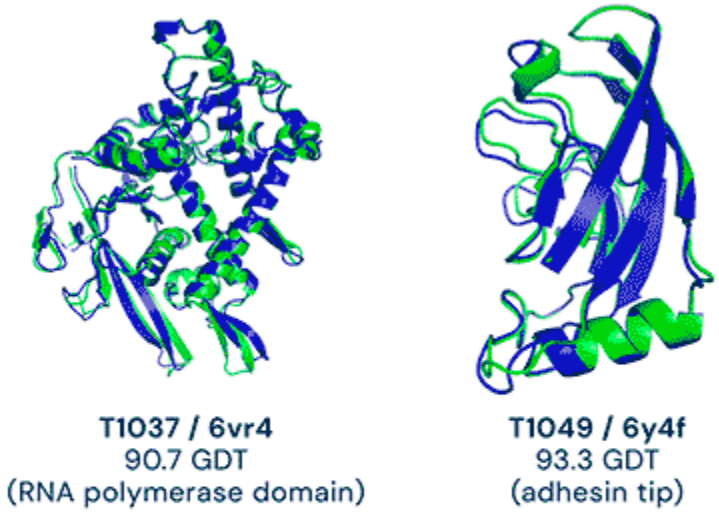
PTM Site Prediction

https://www.musite.net/





Protein Structure Prediction



https://alphafoldserver.com/

THANK YOU!